FYS2160Lab1

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September 27, 2023

Abstract

In this report, we aim to understand the behavior of ideal gases under different conditions of temperatures and estimate the deviation from the theoretical values.

I - Introduction

Thermodynamics focuses on understanding the thermodynamic behavior of objects. In this report, we are going to investigate the behavior of gases under different conditions of temperature, simultaneously investigating how the speed of sound in such gases will be affected by the behavior of those gases..

II - Molecular dynamic simulations

In this task we want to use molecular dynamics simulations to determine the degrees of freedom f of atoms and molecules. We use Lammps to calculate changes in energy U, temperature T and pressure P, and use the thermodynamic relations regarding heat capacity C_V and compressibility Z to estimate the degrees of freedom. Lastly, we compare the results to that of the ideal gas.

Since Lammps is returning dimensionless units, we need to define some new notation:

$$U^* = \frac{U}{\epsilon} \tag{1}$$

$$T^* = \frac{kT}{\epsilon} \tag{2}$$

$$P^* = \frac{P}{\epsilon}. \tag{3}$$

To calculate the degrees of freedom, we will need the heat capacity

$$C_V = \left(\frac{\partial U}{\partial T}\right) \tag{4}$$

and to determine the number of degrees of freedom we use the equipartiion theorem

$$U = \frac{1}{2} f N k T \tag{5}$$

$$U^* \epsilon = \frac{1}{2} f N k \frac{T^* \epsilon}{k} \tag{6}$$

$$U^* = \frac{1}{2} f N T^* \tag{7}$$

$$f = \frac{2U^*}{NT^*} \tag{8}$$

$$f = \frac{2E_{tot}}{T^*} \tag{9}$$

$$f = 2C_V. \tag{10}$$

We also want to calculate the compressibility, which for an ideal gas $Z = 1$, and in general

$$Z = \frac{PV}{NkT} = \frac{P^*}{T^* \rho} \tag{11}$$

where we have used $\rho = \frac{N}{V}$.

III - Experiment

The aim of was to recreate *August Kundt's* experiment, where he looked at standing waves inside what he called *Kundt's tube*, with modern equipment and test thermodynamic gas theory.

We know that sound is a longitudinal pressure wave, and the speed of this propagating wave depends on a variety of thermodynamic quantities and parameters; such as temperature, pressure, and the slope a from a linear regression of the resonance frequencies. To determine the speed of sound in a gas, we start by identifying the resonance frequencies of standing waves inside the tube. By combining the knowledge that the velocity of a wave, c, is given as $c = \lambda\nu$, and that the wavelength of a standing wave inside a closed tube is given as $\lambda_n = 2L/n$, we obtain the following expression for the speed of sound:

$$\boxed{\nu_n = an + b, \quad a = \frac{c}{2L}} \tag{12}$$

where a is the slope of a linear regression done on values of U^* and T^*.

Carrying out the experiment entailed dividing up the tasks between all three participants; one participant wrote down the measurements by hand, while the second and third implemented them in code and managed the oscilloscope respectively.

IV - Method and Results

Speed of sound

To determine the speed of sound in difference gasses, we measure the resonance frequency of a signal through the gasses in a tube, and we use the relation

$$\nu_n = an + b, \quad a = \frac{c}{2L} \tag{13}$$

where ν_n is resonance frequency number n, c is the speed of sound we are looking for, L is the length of the tube and a and b are parameters we can find from linear regression.

The value of c from this model can be compared to that of the ideal gas model

$$c = \sqrt{\frac{(f+2)RT}{fM_{mol}}} \tag{14}$$

where f is the number of degrees of freedom, R is the ideal gas constant and M_{mol} is the molar mass. By finding the speed of sound for a gas at three different temperatures, we can use this formula to check if it matches with the degrees of freedom with the relation

$$\frac{f+2}{f} = \frac{dM_{mol}}{R} \tag{15}$$

where d is found as the slope of $(c(t))^2$.

We had four tubes (K1, K2, K3, and K4) each with a length of

$$1243\,\text{mm} = 1.243\,\text{m} \pm 1.5\,\text{mm}$$

The tube K1 contains argon or CO2 at $T = T_{\text{room}}$C, K2 contains air at room temperature, K3 containg air at $T \approx 70°$ and K4 containing air at $T \approx 50°$.

From the frequency measurements with the K1 tube with air, we get the results in table 1 which is illustrated in FIG 2 for argon and FIG 3 for CO2. Calculating the slope a from these we get $c = 311 \pm 4$ m/s for argon and $c = 267 \pm 1$ m/s for CO2, which is close to the theoretical 323 m/s and 258 m/s, respectively. Therefore, we can say that using $f = 3$ for argon and $f = 5$ for CO2 seems to be a good fit, although we don't hit the theoretical values exactly, not even within the uncertainty.

The measurements for the resonance frequencies for air at different temperatures in tubes K2, K3 and K4 are given in table 2 which is illustrated in the FIGs 4, 5 and 6. Calculating c, i.e. the slopes, from these, we get the speeds $c_{24} = 338 \pm 1$ m/s, $c_{47} = 367 \pm 1$ m/s and $c_{66} = 369 \pm 1$ m/s, where the subscript indicates the temperature in Celsius.

By using linear regression we try to determine the degrees of freedom from equation **??**, but without getting reasonable results. However, as for $c = 338 \pm 1$ m/s at room temperature, with the theoretical value being about 340 m/s, we can say that our choice of $f = 5$ for air seems to be accurate at least for this temperature.

Table 1: Resonance frequencies measured in Hz for argon and CO2 in the K1 tube.

Argon	CO2
204	224
269	329
392	437
820	544
644	651
779	758
905	865
1034	972
1163	1081
1294	1190
1423	1297
	1407
	1511
	1621

V - Tasks

Task 1

We start by running the simulation from the input file `heatcaplj.in`. We reduce the density to 0.001 to ensure a gaseous phase, and we increase the number of iterations to 10^5 to make the output smoother.

The simulation gives the energy-temperature relation shown in FIG ??. From this we use `scipy.stats.linregress` in python to determine the slope of this graph, i.e. the heat capacity from eq. **??**. We estimate the compressibility from eq: 11 by using the same method.

Calculating this gives a heat capacity of $C_V = 1.4904 \pm 0.0004$ which from eq. 5 gives $f = 2.98 \approx 3$ degrees of freedom. The compressibility becomes $Z = 0.997 \approx 1$.

As we for a single atom in an ideal gas would expect three degrees of freedom and a compressibility of one, we see that our computations are not far off. Running the code developed for this task, yields:

```
From the simulation for Task 1:
C_V: 1.49045 ± 0.0004
f: 2.9809
Z: 0.99701 ± 0.000002
```

Task 2

In this task we repeat the analysis done in task **1.**, the sole difference being that instead of running multiple iterations with increasing temperature, we instead modify the input such that the temperature yields a large increase within a single iteration.

Using our knowledge from the previous task, we get the following output:

Table 2: Resonance frequencies measured in Hz for air at three different temperatures.

24.9° C	46.9° C	66° C
295	255	303
473	440	460
273	586	607
563	736	756
701	882	907
840	1027	1056
979	1174	1207
1116	1319	1357
1257	1456	1507
1397	1612	1817
1537	1759	1959
1676	1903	
1815		
1956		

```
From the simulation for Task 1:
C_V: 1.49045 ± 0.0004
f: 2.9809
Z: 0.99701 ± 0.000002

From the simulation for Task 2:
C_V: 1.49726 ± 0.0000
f: 2.99453
Z: 0.999163 ± 0.000000
```

From the output we draw the conclusion that the heat capacity of a gas at a fixed volume, C_V, and the sum of the Boltzmann factors, Z, are not temperature dependent - or rather, that they are to a very small degree temperature dependent.

Task 3

Now, using the configuration for task **1** as our baseline, the aim is to repeat the simulations for increasing densities. Since the density is defined at the beginning of the simulation, one test is performed per density. Since our fluid undergoes phase changes at $\rho \geq 0.85$, we have chosen to run simulations for $\rho \in (0.084, 0.84)$. To do this, ten different **.in**-files were needed, with different values of ρ, so that we could output out-files for each ρ. Since we found it too tedious to copy the **.in**-file and manually change ρ, which would take at least fifteen to twenty-five minutes, we instead opted to spend two hours developing a C++ and python script to automate the process.

From Fig. (7) and Fig. (8), we see that both the heat capacity C_V and compressibility Z correlate with the density of the gas, ρ.

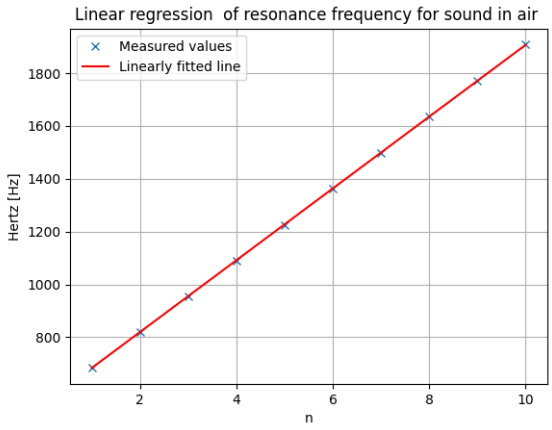


Figure 1: Resonance frequencies of sound in air at room temp. Slope of the linreg of datavals for air: 135.982

Task 4

In this task, we perform a similar simulation to the previous tasks with N_2 molecules. The aim here is to determine the degrees of freedom for the molecule, so the density was set at a low, fixed value for ease of purpose. We quickly noted that the initial value set for the temperature was not adequate, and did not cover a suitable temperature range. Upon altering it from $T = 1$ to $T = 1.2$, we noticed that the plots became more reasonable - albeit there was one mishap.

```
======= Rod =======

From simulations:
C_V: 1.40039 ± 0.0057
f: 2.80079
Z: 1.10726 ± 0.000241

======= Spring =======

From simulations:
C_V: 1.86041 ± 0.0137
f: 3.72082
Z: 1.39059 ± 0.000560
```

From the output, we can see that there is an inaccuracy for the value f. Upon closer inspection, it became evident that it was the choice of the density ρ in the **.in**-file that altered the values of f greatly. We expected that the diatomic rod would have 5 degrees of freedom, and that the diatomic spring would have a few more, and upon changing the value for ρ to be ≈ 0.1, we got our desired values. However, this would change the scope of the task, and alter parameters that were not meant to be altered - so we kept it the same.

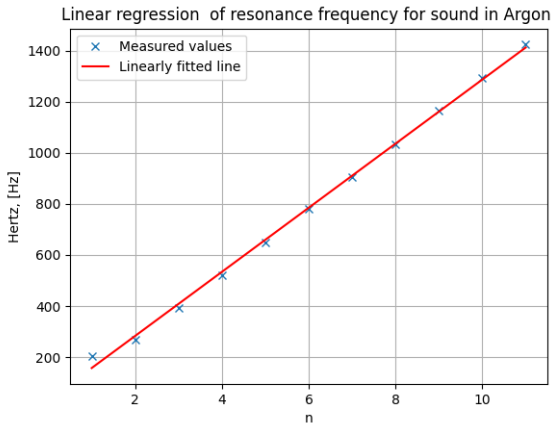


Figure 2: Resonance frequencies in argon, at room temp. Slope of the linreg of datavals for argon:125.4

References

[1] Daniel V. Schroeder. *An Introduction to Thermal Physics*. Addison Wesley Longman, 2000.

[2] Anders Malthe-Sorenssen. *Elementary Thermal and Statistical Physics Using Python*. Springer, 2016.

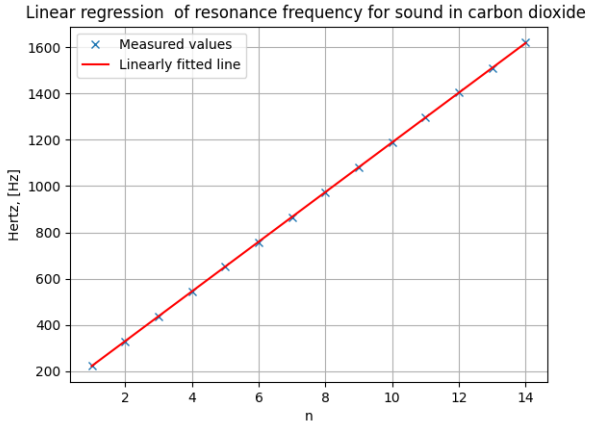


Figure 3: Resonance frequencies of sound in Carbon dioxide, at room temp. Slope of the linreg of datavals for carbon dioxide:107.549

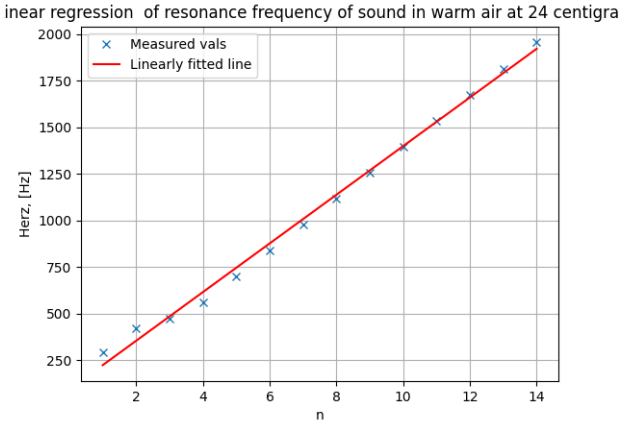


Figure 4: Resonance frequencies of sound in air at a 24 centigrade. Slope of the linreg of datavals for air in 24 centigrade:130.589 ± 2.67676

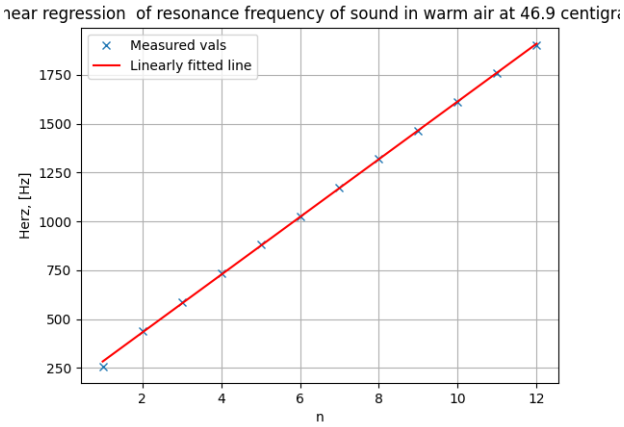


Figure 5: Resonance frequencies in air at 46.9 centigrade. Slope of the linreg of datavals for air in 47 centigrade:147.846 ± 2.67676

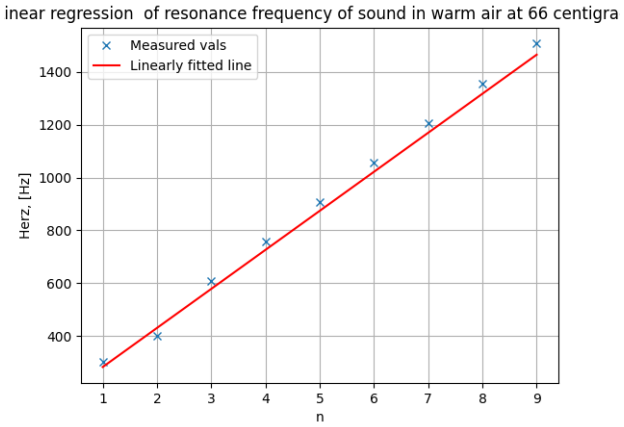


Figure 6: Resonance frequencies in air at 66 centigrade. Slope of the linreg of datavals for air in 66 centigrade: 153.11 ± 2.67676

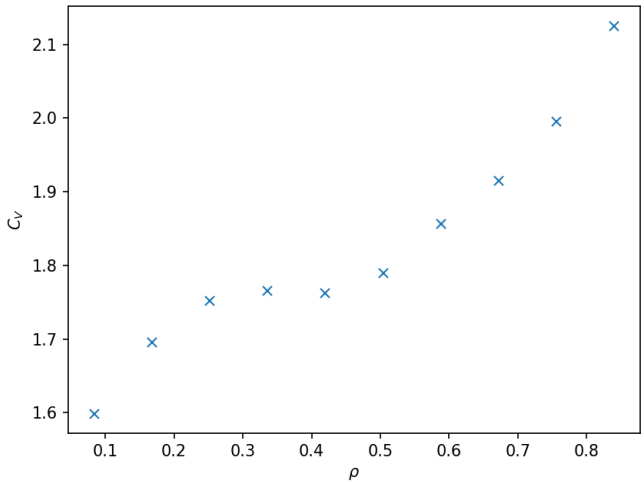


Figure 7: Plot showing the correlation between the heat capacity C_V and density ρ.

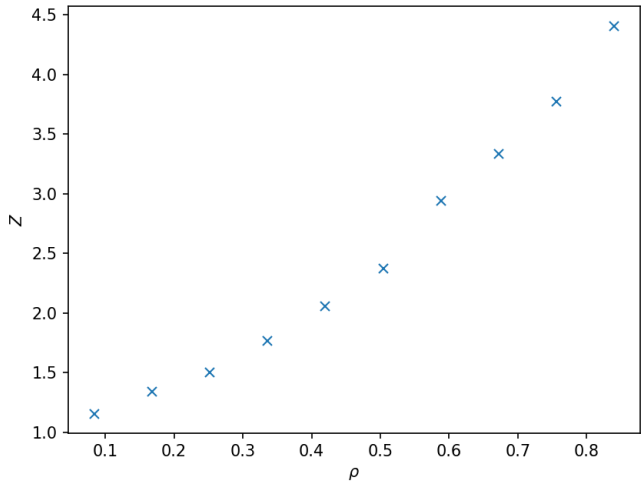


Figure 8: Plot showing the correlation between the compressibility Z and density ρ.

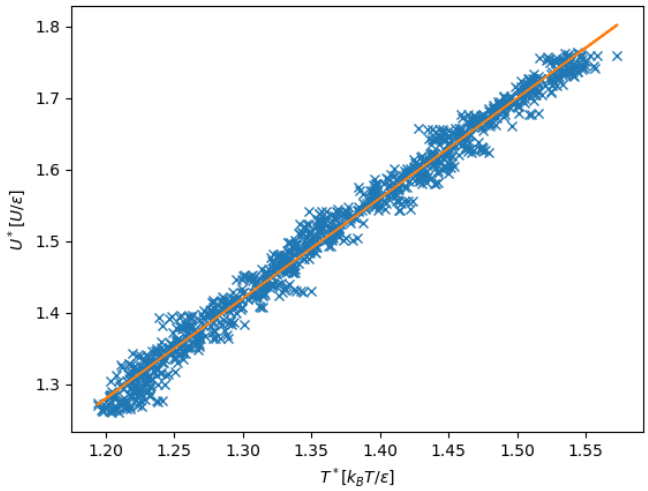


Figure 9: Plot of measured values of U^* and T^* for a diatomic rod, and a fitted line found by doing a linear regression on the measured values.

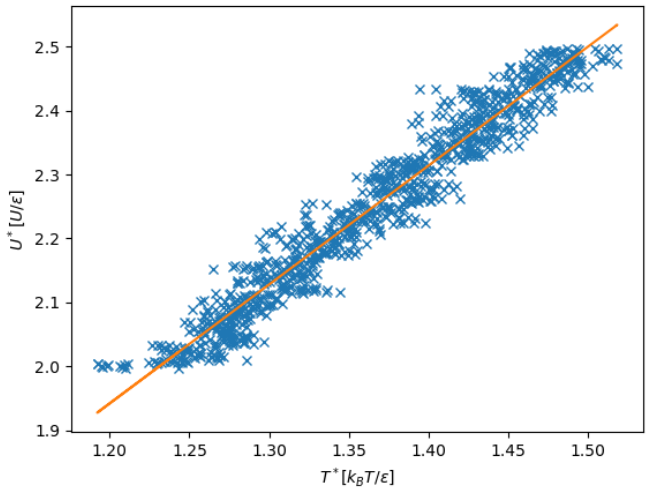


Figure 10: Plot of measured values of U^* and T^* for a diatomic spring, and a fitted line found by doing a linear regression on the measured values.